

Problems**Example 8.13**

Consider the program for the determination of the nature of roots of a quadratic equation. Its input is a triple of positive integers (say a, b, c) and values may be from interval $[0, 100]$.

The program is given in Fig. 8.19. The output may have one of the following words: [Not a quadratic equation; real roots; Imaginary roots; Equal roots]

Draw the flow graph and DD path graph. Also find independent paths from the DD path graph.

```

#include <stdio.h>
#include <conio.h>
#include <math.h>
1   int main()
2   {
3       int a,b,c,validInput=0,d;
4       double D;
5       printf("Enter the 'a' value: ");
6       scanf("%d",&a);
7       printf("Enter the 'b' value: ");
8       scanf("%d",&b);
9       printf("Enter the 'c' value: ");
10      scanf("%d",&c);
11      if ((a >= 0) && (a <= 100) && (b >= 0) && (b <= 100) && (c >= 0)
        && (c <= 100)) {
12          validInput = 1;
13          if (a == 0) {
14              validInput = -1;
15          }
16      }
17      if (validInput==1) {
18          d = b*b - 4*a*c;
19          if (d == 0) {
20              printf("The roots are equal and are r1 = r2 = %f\n",
                -b/(2*(float) a));
21          }
22          else if ( d > 0 ) {
23              D=sqrt(d);
24              printf("The roots are real and are r1 = %f and r2 = %f\n",
                (-b-D)/(2* a), (-b+D)/(2* a));
25          }
26          else {
27              D=sqrt(-d)/(2*a);
28              printf("The roots are imaginary and are r1 = (%f,%f) and
                r2 = (%f,%f)\n", -b/(2.0*a),D,-b/(2.0*a),-D);
29          }
30      }
31      else if (validInput == -1) {

```

(Contd.)..

```

32  / printf("The values do not constitute a Quadratic equation.");
33  }
34  else {
35      printf("The inputs belong to invalid range.");
36  }
37  getch();
38  return 1;
39  }

```

Fig. 8.19: Code of quadratic equation problem

Solution

The flow graph is given below:

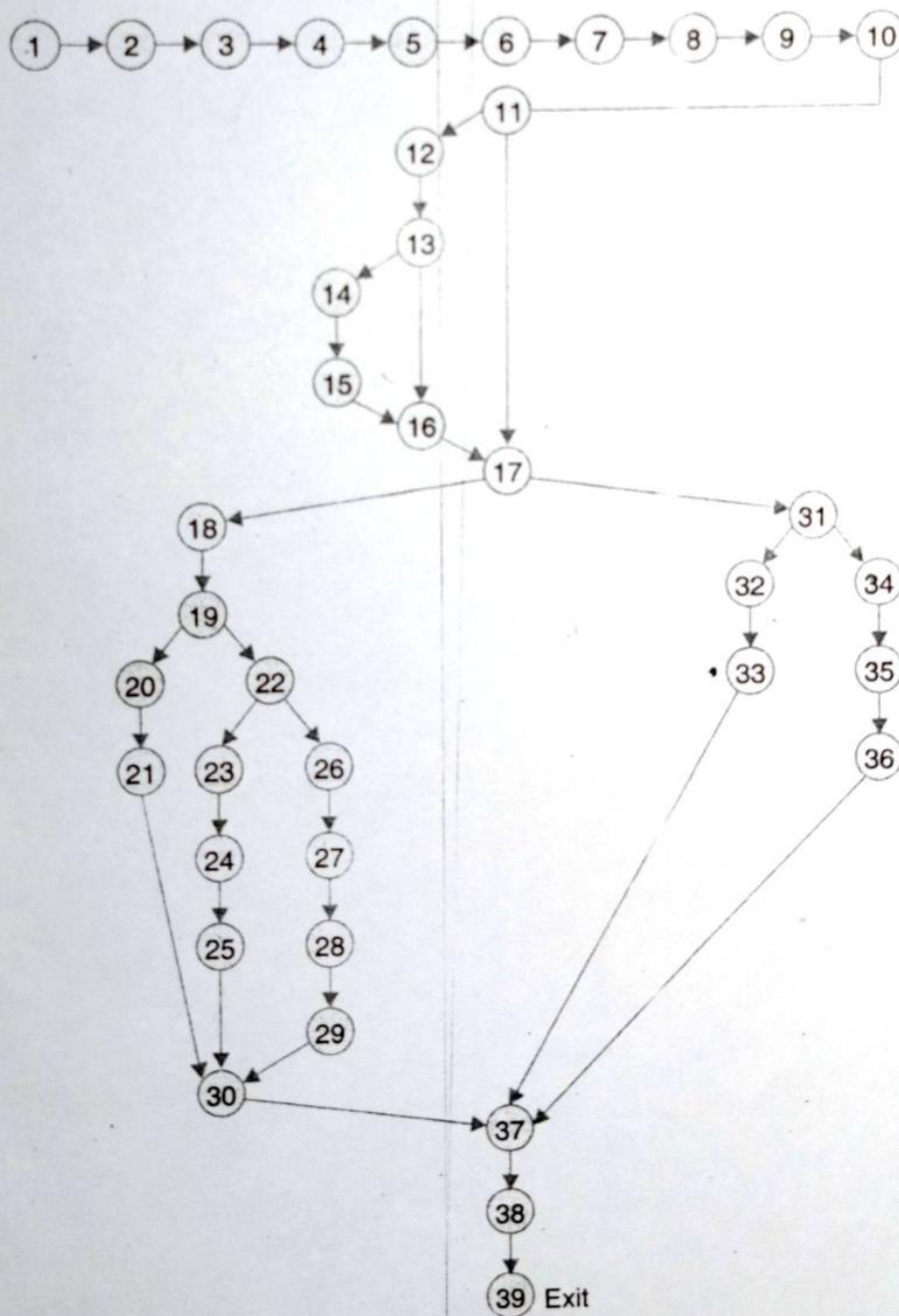


Fig. 8.19(a): Program flow graph

DD Path graph is given below :

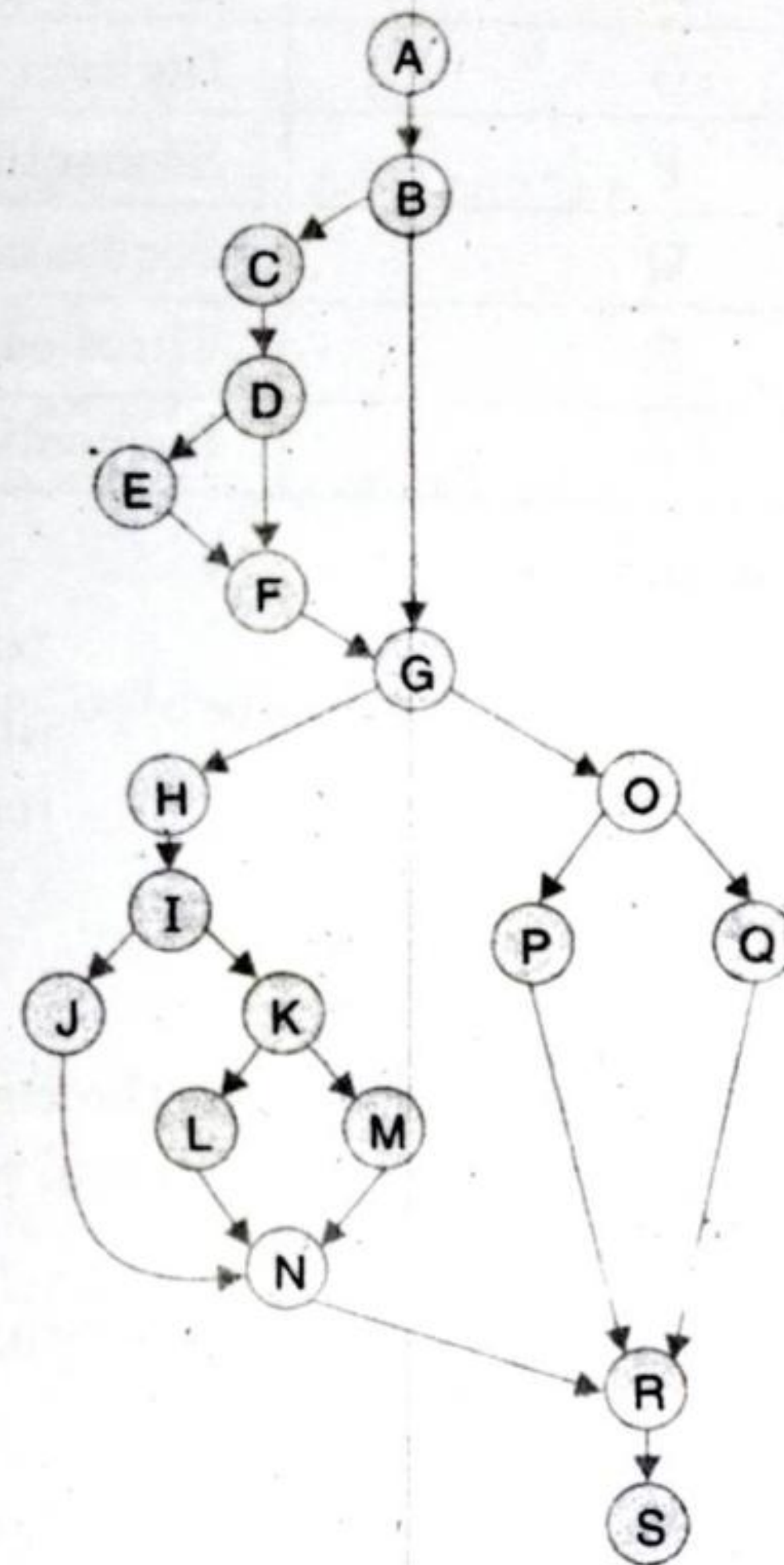


Fig. 8.19(b): DD path graph

The mapping table for DD path graph is:

Flow graph nodes	DD path graph corresponding nodes	Remarks
1 to 10	A	Sequential nodes
11	B	Decision node
12	C	Intermediate node
13	D	Decision node
14, 15	E	Sequential nodes
16	F	Two edges are combined here
17	G	Two edges are combined and decision node
18	H	Intermediate node
19	I	Decision node
20, 21	J	Sequential node
22	K	Decision node
23, 24, 25	L	Sequential nodes
26, 27, 28, 29	M	Sequential nodes

(Contd.)...

30	N	Three edges are combined
31	O	Decision node
32, 33	P	Sequential nodes
34, 35, 36	Q	Sequential nodes
37	R	Three edges are combined
38, 39	S	Sequential nodes with exit node.

Independent paths are:

- | | |
|------------------|------------------|
| (i) ABGOQRS | (ii) ABGOPRS |
| (iii) ABCDFGOQRS | (iv) ABCDEFGOPRS |
| (v) ABGHIJNRS | (vi) ABGHIKLNRS |
| (vii) ABGHIKMNRS | |

Example 8.14

Consider a program given in Fig. 8.20 for the classification of a triangle. Its input is a triple of positive integers (say, a, b, c) from the interval $[1, 100]$. The output may be [Scalene, Isosceles, Equilateral, Not a triangle].

Draw the flow graph and DD path graph. Also find the independent paths from the DD path graph.

```

#include <stdio.h>
#include <conio.h>
1  int main()
2  {
3      int a,b,c,validInput=0;
4      printf("Enter the side 'a' value: ");
5      scanf("%d",&a);
6      printf("Enter the side 'b' value: ");
7      scanf("%d",&b);
8      printf("Enter the side 'c' value:");
9      scanf("%d",&c);
10     if ((a > 0) && (a <= 100) && (b > 0) && (b <= 100) && (c > 0)
        && (c <= 100)) {
11         if ((a + b) > c) && ((c + a) > b) && ((b + c) > a)) {
12             validInput = 1;
13         }
14     }
15     else {
16         validInput = -1;
17     }
18     If (validInput==1) {
19         If ((a==b) && (b==c)) {
20             printf("The trinagle is equilateral");
21         }
22         else if ( (a == b) || (b == c) || (c == a) ) {

```

(Contd.)...

```

23     printf("The triangle is isosceles");
24 }
25 else {
26     printf("The trinagle is scalene");
27 }
28 }
29 else if (validInput == 0) {
30     printf("The values do not constitute a Triangle");
31 }
32 else {
33     printf("The inputs belong to invalid range");
34 }
35 getch();
36 return 1;
37 }

```

Fig. 8.20: Code of triangle classification problem

Flow graph of trinagle problem is:

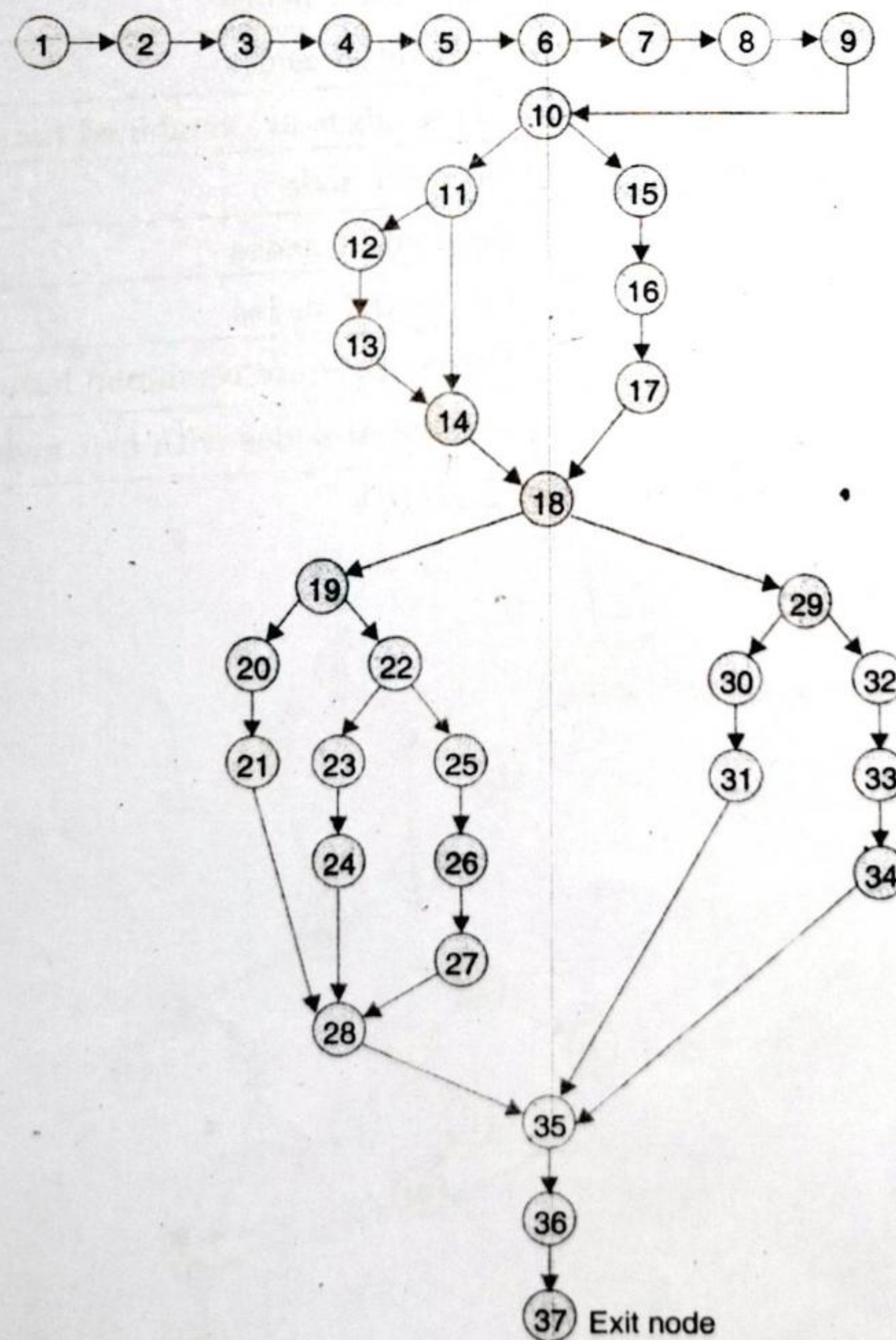


Fig. 8.20(a): Program flow graph

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The mapping table for DD path graph is:

Flow graph nodes	DD path graph corresponding nodes	Remarks
1 to 9	A	Sequential nodes
10	B	Decision node
11	C	Decision node
12, 13	D	Sequential nodes
14	E	Two edges are joined here
15, 16, 17	F	Sequential nodes
18	G	Decision nodes plus joining of two edges
19	H	Decision node
20, 21	I	Sequential nodes
22	J	Decision node
23, 24	K	Sequential nodes
25, 26, 27	L	Sequential nodes
28	M	Three edges are combined here
29	N	Decision node
30, 31	O	Sequential nodes
32, 33, 34	P	Sequential nodes
35	Q	Three edges are combined here
36, 37	R	Sequential nodes with exit node

DD path graph is given in Fig. 8.20 (b).

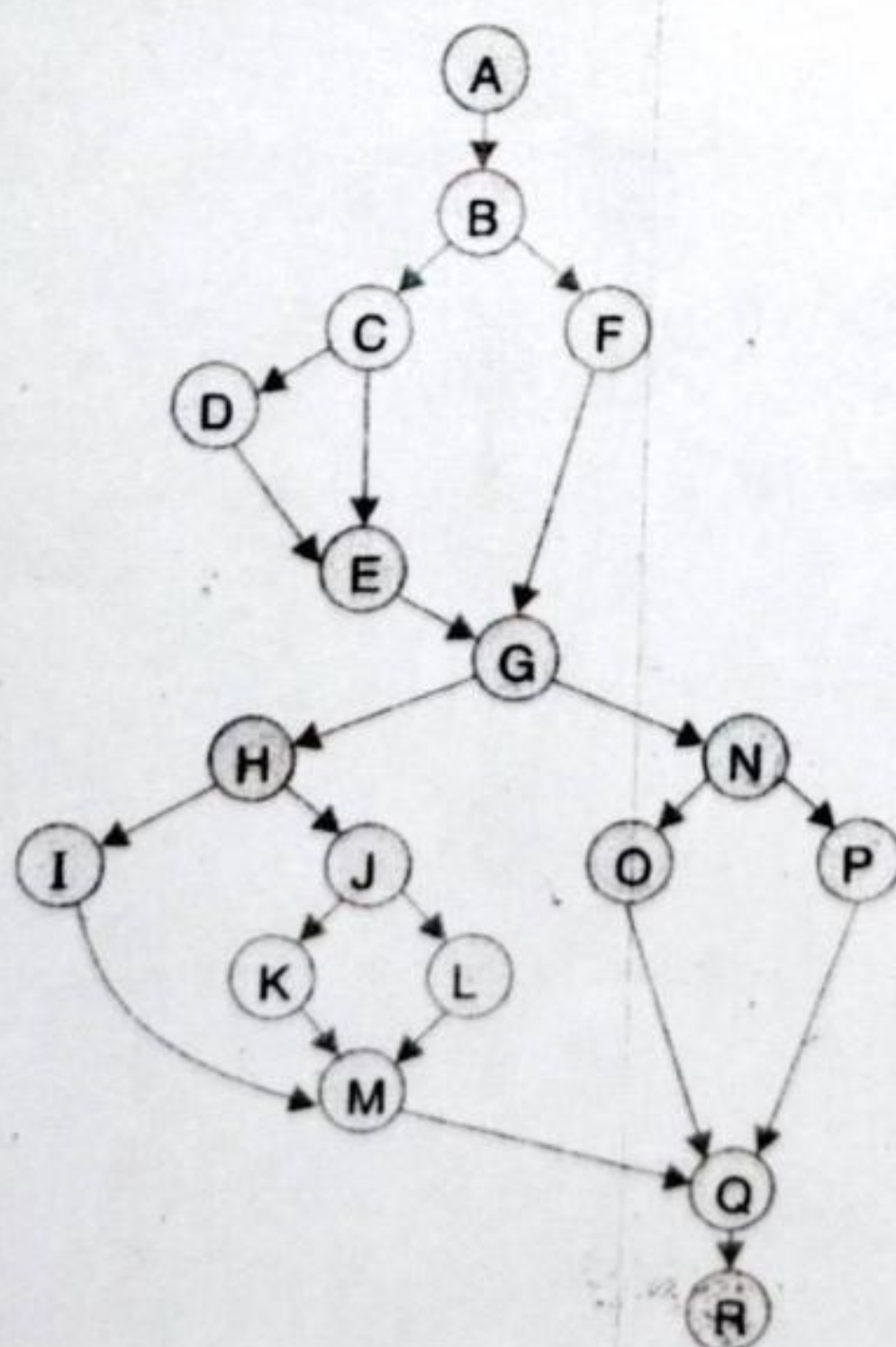


Fig. 8.20(b): DD path graph

Independent paths are:

- | | |
|------------------|-----------------|
| (i) ABFGNPQR | (ii) ABFGNOQR |
| (iii) ABCEGNPQR | (iv) ABCDEGNOQR |
| (v) ABFGHIMQR | (vi) ABFGHJKMQR |
| (vii) ABFGHJLMQR | |